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TASK: MITRE ATT&CK Mapping Exercise

Threat Report: “128 Seconds to disruption: Microsoft Defender stops ransomware at QNET”

Introduction:

This report analyzes the “128 Seconds to Disruption: Microsoft Defender Stops Ransomware at QNET” incident using the MITRE ATT&CK Framework. The objective is to identify the attack tactics and techniques used by the attacker, map them to their corresponding MITRE ATT&CK technique IDs, and understand how the attack progressed. The report also highlights the security detections that could identify each stage of the attack and demonstrates the mapped techniques using the MITRE ATT&CK Navigator.

MITRE ATT&CK Mapping Table:

Attack Stage	MITRE Tactic	Technique	MITRE ATT&CK ID
Malicious file opened by the user.	Initial Access / Execution.	Phishing (file downloaded through email or browser link) / User Execution: Malicious File.	T1566 / T1204.002
mshta.exe was executed by the malicious file.	Stealth.	System Binary Proxy Execution: Mshta	T1218.005
mshta.exe connected to the attacker-controlled URL.	Command and Control.	Application Layer: Web Protocols.	T1071.001
Second-stage payload executed.	Execution.	Command and Scripting Interpreter.	T1059
RunMRU activity was observed.	Persistence.	Modify Registry.	T1112

MITRE ATT&CK Navigator Mapping:

Microsoft Defender Ransomware Incident Mapping															Selection Controls Layer Controls Technique Controls		
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Reconnaissance 12 techniques	Resource Development 9 techniques	Initial Access 11 techniques	Execution 20 techniques	Persistence 22 techniques	Privilege Escalation 13 techniques	Stealth 30 techniques	Defense Impairment 18 techniques	Credential Access 17 techniques	Discovery 34 techniques	Lateral Movement 9 techniques	Collection 17 techniques	Command and Control 18 techniques	Exfiltration 9 techniques	Impact 15 techniques			
Active Scanning 0.1%	Acquire Access 0.1%	Content Injection 0.1%	BITS Jobs 0.1%	Account Manipulation 0.1%	Abuse Execution Control Mechanism 0.1%	Access Token Manipulation 0.1%	Disable or Modify System Firewall 0.1%	Adversary-in-the-Middle 0.1%	Account Discovery 0.1%	Exploitation of Remote Services 0.1%	Adversary-in-the-Middle 0.1%	DNS 0.1%	Automated Exfiltration 0.1%	Account Access Removal 0.1%			
Gather Victim Host Information 0.1%	Acquire Infrastructure 0.1%	Drive-by Compromise 0.1%	Cloud Administration Command 0.1%	BITS Jobs 0.1%	Access Token Manipulation 0.1%	Build Image on Host 0.1%	Brute Force 0.1%	Credentials from Password Stores 0.1%	Application Window Discovery 0.1%	Internal Spearphishing 0.1%	Archive Collected Data 0.1%	File Transfer Protocols 0.1%	Data Transfer Size Limits 0.1%	Data Destruction 0.1%			
Gather Victim Identity Information 0.1%	Compromise Accounts 0.1%	Exploit Public-Facing Application 0.1%	Command and Scripting Interpreter 0.1%	Boot or Logon Autostart Execution 0.1%	Build Image on Host 0.1%	Debugger Evasion 0.1%	Disable or Modify Tools 0.1%	Domain or Tenant Policy Modification 0.1%	Browser Information Discovery 0.1%	Remote Service Session Hijacking 0.1%	Audio Capture 0.1%	Mail Protocols 0.1%	Publish/Subscribe Protocols 0.1%	Data Encrypted for Impact 0.1%			
Gather Victim Network Information 0.1%	Compromise Infrastructure 0.1%	External Remote Services 0.1%	Container Administration Command 0.1%	Boot or Logon Autostart Execution 0.1%	Account Manipulation 0.1%	Delay Execution 0.1%	Domain or Tenant Policy Modification 0.1%	Exploitation for Credential Access 0.1%	Cloud Infrastructure Discovery 0.1%	Remote Services Session Hijacking 0.1%	Automated Collection 0.1%	Web Protocols 0.1%	Exfiltration over Alternative Protocol 0.1%	Data Manipulation 0.1%			
Gather Victim Org Information 0.1%	Develop Capabilities 0.1%	Hardware Additions 0.1%	Container Administration Command 0.1%	Cloud Application Integration 0.1%	Boot or Logon Autostart Execution 0.1%	Deobfuscate/Decode Files or Information 0.1%	Downgrade Attack 0.1%	Exploitation for Defense Impairment 0.1%	Cloud Service Dashboard 0.1%	Browser Session Hijacking 0.1%	Communication Through Removable Media 0.1%	Exfiltration Over C2 Channel 0.1%	Defacement 0.1%	Disk Wipe 0.1%			
Phishing for Information 0.1%	Establish Accounts 0.1%	Phishing 0.1%	ESX Administration Command 0.1%	Compromise Host Software Binary 0.1%	Boot or Logon Autostart Execution 0.1%	Direct Volume Access 0.1%	Exploitation for Defense Impairment 0.1%	Exploitation for Defense Impairment 0.1%	Cloud Service Discovery 0.1%	Remote Services 0.1%	Clipboard Data 0.1%	Content Injection 0.1%	Exfiltration Over Other Network Medium 0.1%	Email Bombing 0.1%			
Query Public AI Services 0.1%	Generate Content 0.1%	Replication Through Removable Media 0.1%	Exploitation for Client Execution 0.1%	Create Account 0.1%	Domain or Tenant Policy Modification 0.1%	Execution Guardrails 0.1%	File and Directory Permissions Modification 0.1%	Input Capture 0.1%	Cloud Storage Object Discovery 0.1%	Replication Through Removable Media 0.1%	Data from Configuration Repository 0.1%	Data Obfuscation 0.1%	Exfiltration over Physical Medium 0.1%	Endpoint Denial of Service 0.1%			
Search Closed Sources 0.1%	Obtain Capabilities 0.1%	Supply Chain Compromise 0.1%	Hijack Execution Flow 0.1%	Create or Modify System Process 0.1%	Domain or Tenant Policy Modification 0.1%	Hide Artifacts 0.1%	Modify Authentication Process 0.1%	Modify Authentication Process 0.1%	Container and Resource Discovery 0.1%	Software Deployment Tools 0.1%	Data from Information Repositories 0.1%	Dynamic Resolution 0.1%	Exfiltration over Web Service 0.1%	Financial Theft 0.1%			
Search Open Technical Databases 0.1%	Stage Capabilities 0.1%	Input Injection 0.1%	Inter-Process Communication 0.1%	Event Triggered Execution 0.1%	Domain or Tenant Policy Modification 0.1%	Hijack Execution Flow 0.1%	Multi-Factor Authentication Interception 0.1%	Multi-Factor Authentication Interception 0.1%	Device Driver Discovery 0.1%	Taint Shared Content 0.1%	Domain Trust Discovery 0.1%	Encrypted Channel 0.1%	Inhibit System Recovery 0.1%	Firmware Corruption 0.1%			
Search Open Websites/Domains 0.1%	Valid Accounts 0.1%	Native API 0.1%	Poisoned Pipeline Execution 0.1%	Implant Internal Image 0.1%	Exploitation for Privilege Escalation 0.1%	Pre-OS Boot 0.1%	Modify Cloud Resource Hierarchy 0.1%	Modify Cloud Resource Hierarchy 0.1%	Local Storage Discovery 0.1%	Log Enumeration 0.1%	Network Service Discovery 0.1%	Network Share Discovery 0.1%	Non-Standard Port 0.1%	Protocol Tunneling 0.1%			
Search Threat Vendor Data 0.1%	Scheduled Task/Job 0.1%	Serverless Execution 0.1%	Shared Modules 0.1%	Software Deployment Tools 0.1%	System Services 0.1%	Scheduled Task/Job 0.1%	Control Panel 0.1%	Control Panel 0.1%	Permitted Command History Logging 0.1%	Unsecured Credentials 0.1%	Query Registry 0.1%	Remote System Discovery 0.1%	Software Discovery 0.1%	System Information Discovery 0.1%			
Search Victim-Owned Websites 0.1%	Malicious Copy and Paste 0.1%	Malicious File 0.1%	Malicious Image 0.1%	Malicious Library 0.1%	Malicious Link 0.1%	Windows Management Instrumentation 0.1%	System Binary Proxy Execution 0.1%	System Binary Proxy Execution 0.1%	System Location Discovery 0.1%	System Network Configuration Discovery 0.1%	System Shutdown/Reboot 0.1%	System Shutdown/Reboot 0.1%	System Shutdown/Reboot 0.1%	System Shutdown/Reboot 0.1%			

Detection Summary For Each Attack Stage:

Attack Stage	Possible Detection
Malicious file opened by the user. (Initial Access / Execution)	Email Security Gateway, Microsoft Defender SmartScreen, Secure Web Gateway.
A malicious file executed mshta.exe. (Stealth)	Microsoft Defender for Endpoint (EDR), Process Monitoring, Application Control (AppLocker/WDAC).
mshta.exe connected to the attacker-controlled URL. (Command and Control)	Firewall, IDS/IPS, DNS Monitoring, Proxy Logs, Network Traffic Monitoring.
Second-stage payload downloaded. (Execution)	Network Detection and Response (NDR), Web Proxy Monitoring.
RunMRU activity was observed. (Persistence)	Registry Monitoring, Windows Event Logs (Sysmon), EDR, SIEM Correlation Rules.

Incident Summary:

The incident began when a user opened a malicious file delivered through email or a browser download. The file executed mshta.exe, a legitimate Windows utility that was abused to connect to an attacker-controlled URL and execute a second-stage payload. The attacker then attempted to establish persistence through RunMRU-related registry activity. Microsoft Defender detected the malicious activity and automatically isolated the compromised device within 128 seconds, successfully containing the attack before any lateral movement or significant impact on the organization could occur. This incident also highlights that the attack could have been detected and mitigated at different stages using security controls such as email security gateways, Microsoft Defender SmartScreen, endpoint protection (EDR), firewall and network traffic monitoring, registry monitoring, and SIEM-based log monitoring. Using the MITRE ATT&CK Framework, the attack tactics and techniques were successfully mapped, providing a structured understanding of the attack lifecycle and helping identify appropriate detection opportunities. Overall, this incident demonstrates the importance of layered security controls, continuous monitoring, and rapid incident response in minimizing the impact of modern cyberattacks.